

Tredir Sewpaul

Computer Science Graduate · Software Engineering Enthusiast · AI & Machine Learning

Cape Town, Western Cape, South Africa

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// Profile Summary

I am a Computer Science graduate from the University of the Western Cape, passionate about software engineering, artificial intelligence, and scalable systems. My academic journey and practical experiences have strengthened my technical, analytical, and collaborative skills. I thrive in environments that merge creativity, technology, and teamwork – always seeking to build solutions that make a difference.

With hands-on experience across full-stack web development, machine learning, cross-platform mobile applications, AI engineering, and academic research, I bring both depth and breadth to engineering teams. Whether leading architecture decisions, tutoring students, or competing in hackathons, I consistently demonstrate adaptability, initiative, and a drive to deliver quality work.

// Education

Bachelor of Science in Computer Science

University of the Western Cape · 2018 – 2024

- **Major:** Computer Science
- **Minor:** Information Systems
- **Elective:** Physics
- Merit Award in Mathematics (MAM152) – 2021
- Merit Award in Astrophysics (PHY217) – 2023

Samsung Future Innovation Lab – Continuing Education

University of the Western Cape / Samsung South Africa / Microsoft SA · March 2025 – December 2025

A university-certified industry programme focused on building production-grade digital solutions:

- **CES143** – Digital Social Innovation for Impact
- **MPS** – Multi-Platform Solutions Development

Technical Focus: .NET, C#, .NET MAUI, AI-assisted development (GitHub Copilot, ChatGPT), cloud integration, Agile, Scrum, and Design Thinking aligned with the UN Sustainable Development Goals

(SDGs).

Delivered end-to-end digital solutions through industry-moderated projects.

Matric – Information Technology

Stanger Manor Secondary · 2012 – 2017

// Professional Experience

AI Engineering Intern

Department of Pharmacy, University of the Western Cape · December 2025 – Present

Technical lead guiding architecture across multiple AI research streams:

- Lead architecture across supervised ML, LLM reasoning/summarisation, and vector search pipelines
- Built document similarity models with evidence-linked outputs and human override paths
- Implemented explainability, traceability, and governance-aligned controls
- Managed Azure-based experimentation, ingestion, and model versioning environments
- Led weekly technical reviews and aligned cross-team delivery toward responsible AI milestones

Lead Physics Tutor

University of the Western Cape · February 2020 – 2024

- Supervised and guided 20–30 students per first-year Physics (PHY151) lab session
- Planned, prepared, and delivered experimental activities and lab materials
- Graded and provided detailed feedback on reports and examinations
- Collaborated with fellow tutors for lesson planning and academic support programmes

Website Developer

Meals on Time, Kraaifontein · January 2022 – December 2024

- Developed and maintained the company's informational website showcasing products and menus
- Created and managed product catalogues with live content updates
- Handled social media presence and ongoing website functionality improvements

Website Administrator & Developer

Cuban Connection, Goodwood · November 2022 – December 2023

- Built and managed a full e-commerce platform (www.cigarsemporium.com)
- Handled IT-related tasks including hardware and software maintenance
- Created and maintained online product catalogues
- Managed customer engagement, online sales, and social media interactions

Mathematics & English Tutor

LM Tutoring, Kraaifontein · June 2023 – December 2024

- Tutored Grade 6–11 students in Mathematics and English
- Developed personalised lesson plans and custom teaching materials
- Assessed student progress and adapted teaching methods to individual learning needs

Computer Science Volunteer

UWC IT Society & Women in Tech · April 2024 – Present

- Visited high schools across Cape Town to promote STEM careers
- Delivered presentations on Computer Science pathways and tertiary education opportunities
- Organised hands-on activities to engage students in technology and critical thinking

// Technical Skills

Programming Languages

Python · C# · Java · MATLAB

Web Development

HTML · CSS · JavaScript · PHP · SQL

Frameworks & Tools

.NET MAUI · ASP.NET Core · React · Vite · Git · GitHub · SignalR · Entity Framework Core

Cloud & AI

Azure OpenAI · Azure Cognitive Services · Azure Speech · Azure Translator · Azure Blob Storage · GitHub Copilot

Methodologies

Agile · Scrum · Design Thinking · Responsible AI · Test-Driven Development

Software

Microsoft Word · Excel · PowerPoint

Soft Skills

Problem Solving · Analytical Thinking · Communication · Team Collaboration · Adaptability · Leadership · Mentoring

// Awards & Honours

Award	Year
TelkomX Hackathon Winner – AI-assisted call centre solution	August 2025
3rd Place – South African Intersivity Hackathon · AI Category (FinanceBuddy)	September 2025

Merit Award in Astrophysics (PHY217)	2023
Merit Award in Mathematics (MAM152)	2021

// Co-Curricular Activities

Standard Bank Hackathon – UniHack

June 2024

Collaborated in a 5-member team to develop and prototype a transport industry solution focused on community-driven innovation and sustainable mobility.

SANReN Cyber Security Challenge (CSC)

October 2024

Participated in a national cybersecurity competition covering:

- Capture the Flag (CTF)
- Penetration Testing
- Digital Forensics
- Cryptography
- Malware Analysis & Reverse Engineering

Collaborated with teammates to defend systems and solve complex adversarial scenarios against live simulated cyberattacks.

// Portfolio Projects

Full interactive project showcases are available at www.iamtredir.com/projects

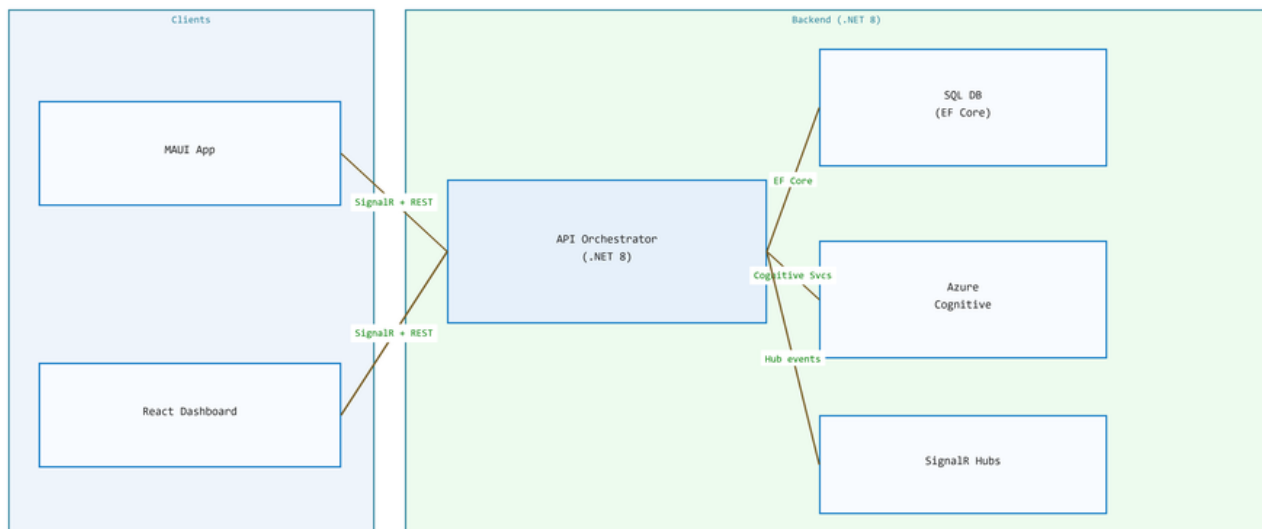
Telkom AI Call Centre Management Solution

TelkomX Hackathon Winner · August 2025 · Ongoing

GitHub: github.com/DrVanHelsing

A comprehensive, multi-project AI-assisted call centre platform combining real-time supervision, cross-platform mobile clients, and NLP-powered analytics. Built as the winning solution for the TelkomX Hackathon.

Architecture:



Key capabilities: - Real-time session monitoring and lifecycle management via SignalR - AI-assisted response generation and conversation orchestration - Text analytics: sentiment analysis, key phrase extraction, named entity recognition, conversation summarisation - React supervisor dashboard with live updates, role-based controls, and offline settings - Cross-platform .NET MAUI agent app (Android, iOS, Windows, Mac Catalyst)

Technology Stack: ASP.NET Core (.NET 8/9) · .NET MAUI (.NET 9) · React (Vite) · SignalR · Azure Cognitive Services · Entity Framework Core · C# · TypeScript

StudentLink – AI-Powered Career Readiness Platform

April 2025 · Ongoing

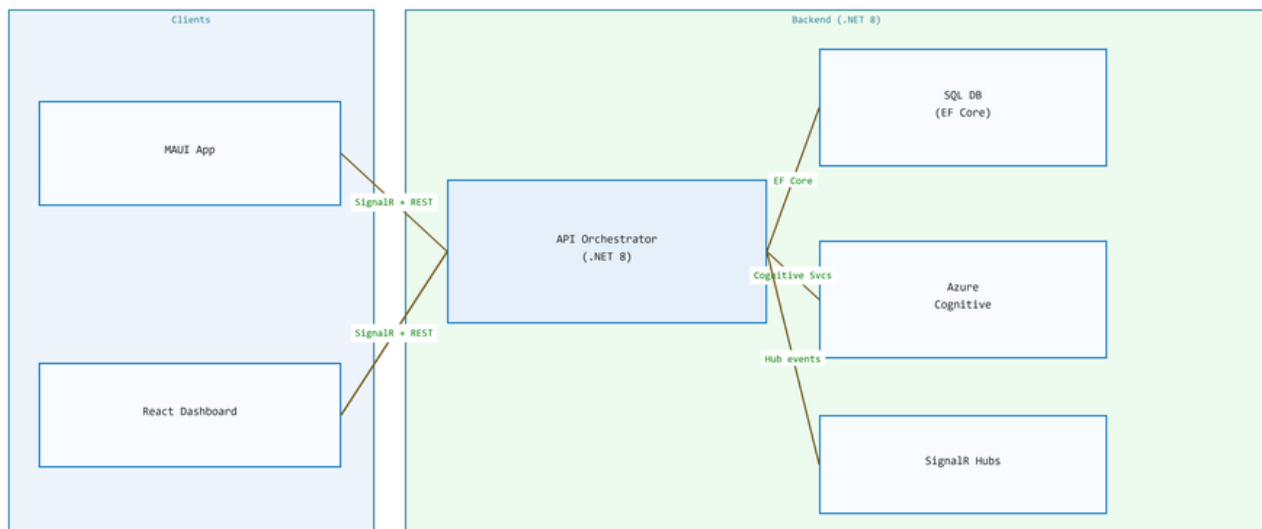
A comprehensive multi-platform career readiness solution connecting students with job opportunities through AI-driven CV analysis, personalised recommendations, and streamlined application workflows.

Components: - **StudentLinkApi** – ASP.NET Core Web API (.NET 9) backend - **StudentLinkFrontend** – React web application for students, recruiters, and admins - **StudentLinkApp** – .NET MAUI mobile app (iOS, Android, Windows) - **StudentLinkSharedLibrary** – Shared models and utilities

Key features: - AI-powered CV analysis with section-by-section feedback, improvement suggestions, and quality scoring - ML-driven job matching using embeddings and feature engineering - Role-based access control: Student / Recruiter / Admin - Recruiter portal for posting, managing, and tracking applications - Visual progress tracking metrics for CV improvement over time - Gamified learning paths and interactive feedback with before/after examples

Technology Stack: ASP.NET Core (.NET 9) · .NET MAUI · React · Azure OpenAI · Azure Document Intelligence · Azure Blob Storage · SQL Server · Entity Framework Core · C# · JavaScript

Architecture:



FinanceBuddy – AI Financial Wellness App

3rd Place · SA Intervarsity Hackathon (AI Category) · September 2025

GitHub: github.com/DrVanHelsing/SAIntervarsityHack2025-MoneyMentor

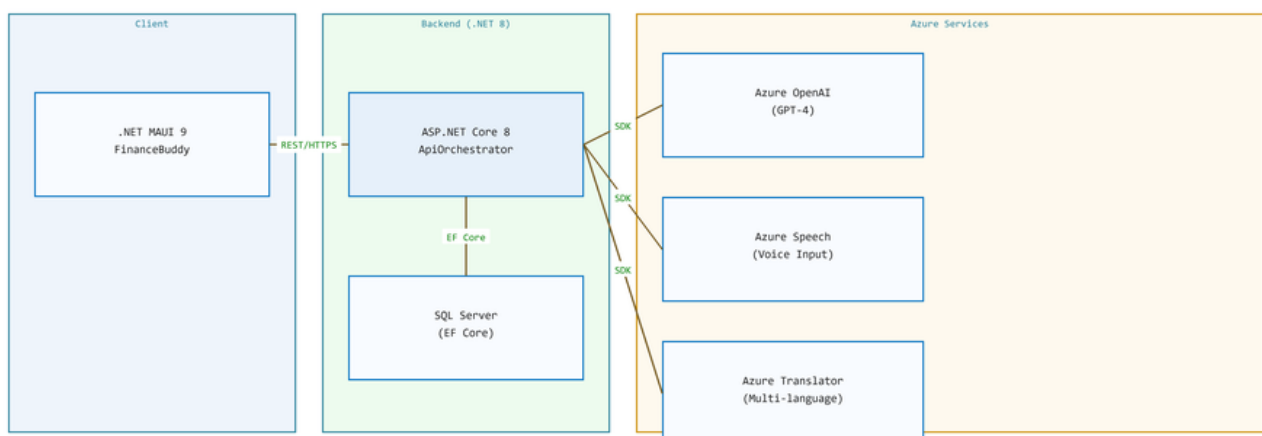
An AI-powered cross-platform mobile app combining intelligent expense tracking, Azure OpenAI-powered financial coaching, multi-language support, and a unique plant-based gamification system to build lasting financial health habits.

Gamification system – your plant grows through 6 wellness stages: **Seed (0)** → **Sprout (100)** → **Seedling (300)** → **Young Plant (600)** → **Mature Plant (1000)** → **Blooming Tree (1500+)**

Key features: - Natural language AI Money Mentor powered by Azure OpenAI GPT-4 - Voice input and multi-language support via Azure Translator - Automatic expense categorisation with visual analytics - Real-time gamification – points for logging, goal-setting, streaks, and weekly reviews - Cross-platform sync between devices

Technology Stack: .NET MAUI 9.0 · ASP.NET Core 8.0 · Azure OpenAI (GPT-4) · Azure Speech · Azure Translator · Entity Framework Core · SQL Server · C#

Architecture:



April 2025

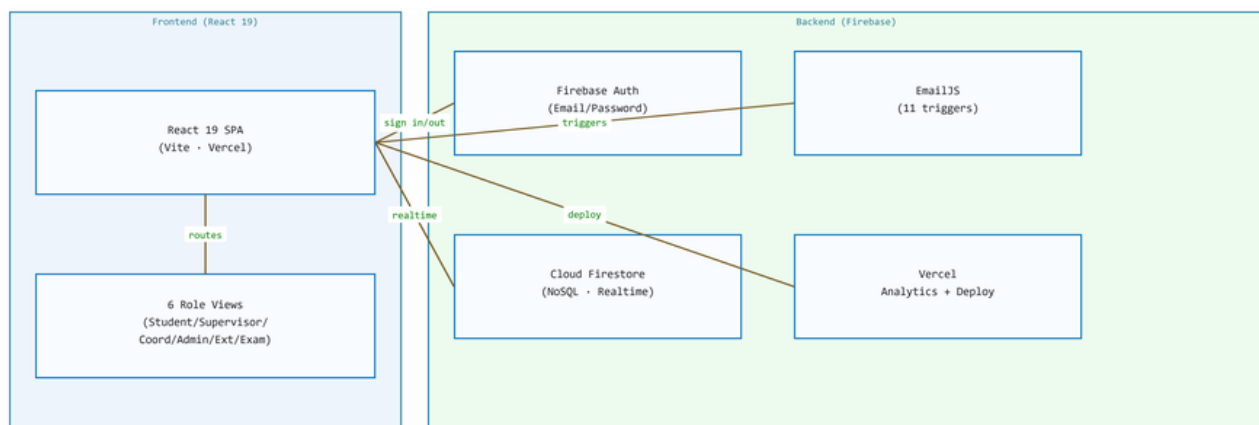
A role-based postgraduate request management system for the University of the Western Cape. Built with React 19, Vite, Firebase, and deployed on Vercel.

Six user roles: Student · Supervisor · Coordinator · Admin · External Examiner · Examiner

Key capabilities: - Full HD request lifecycle: Draft → Submitted → Co-Supervisor → Coordinator → Faculty Board → Senate Board → Approved - 20 prebuilt HDR form templates (HDR-001 to HDR-020) with 15 dynamic field types - Inline annotation system: field-level comments, section threads, text-highlight annotations - Admin form builder with visual drag-and-drop editor and live preview - Multi-version document management with PDF viewer and inline highlight annotations - Real-time notifications with EmailJS integration (11 email trigger functions) - Digital signatures, DOCX export, committee CSV exports, audit logs - Full dark/light mode with persistent preference - Guided tour system with 4 role-specific full-system walkthroughs

Technology Stack: React 19 · Vite · Firebase Auth · Cloud Firestore · Vercel · EmailJS · JavaScript · CSS

Architecture:



GodsEye – Gesture-Controlled Gaming Mouse

April 2025

GitHub: github.com/DrVanHelsing/GodsEye

An experimental prototype for low-latency, contactless cursor control and gaming input using facial gestures, head pose estimation, eye gaze, and blink detection – designed for both gaming and accessibility scenarios.

Key features: - Head pose estimation for smooth cursor movement with configurable sensitivity - Blink-to-click mapping (short blink = left click / long blink = right click) - Gaze-driven cursor movement with smoothing and filtering - Extensible gesture pipeline supporting multiple CV backends (OpenCV, MediaPipe, dlib) - Per-game profile mappings and configurable gesture bindings

Technology Stack: Python · OpenCV · MediaPipe · NumPy · PyAutoGUI

Machine Learning – Neural Networks & Supervised Learning

University of the Western Cape · CSC312 Assignments

GitHub: github.com/DrVanHelsing/MachineLearning

A series of hands-on assignments implementing foundational supervised learning algorithms from scratch in Jupyter Notebooks:

- **Assignment 1 – Linear Regression:** Hypothesis functions, MSE cost, gradient descent, polynomial feature expansion, feature normalisation, under/overfitting analysis
- **Assignment 2 – Logistic Regression:** Sigmoid activation, cross-entropy cost, decision boundary visualisation, regularisation
- **Assignment 3 – Neural Networks:** Forward propagation, backpropagation, weight initialisation, multi-class classification, hyperparameter experimentation

Technology Stack: [Python](#) · [NumPy](#) · [Matplotlib](#) · [scikit-learn](#) · [Jupyter Notebooks](#)

Structural Geology Field Mapping Simulator

Browser-Based 3D Educational Tool

A browser-based 3D interactive geology simulator for education, training, and research. Features a procedurally generated 2 km × 2 km terrain with geologically accurate stratigraphic layers, hydraulic erosion, real-time water rendering, and a full suite of virtual field instruments.

Highlights: - 148K-vertex procedural terrain carved by 38,000 hydraulic erosion iterations - 6 stratigraphic layers (Precambrian to Quaternary) with realistic fault and fold structures - Unified PBR water rendering: Gerstner waves, Fresnel-Schlick, subsurface scattering, caustics - Virtual field instruments: borehole drilling, strike & dip measurement, cross-section generation, elevation sampling - 9,000+ instanced vegetation objects, 600+ riparian plants, full farm compound with windmill - Physical atmospheric scattering sky dome with animated clouds and Rayleigh/Mie scattering

Technology Stack: [Three.js](#) · [WebGL](#) · [JavaScript](#) · [GLSL](#) · [Node.js](#)

Physics IDE – Visual Physics Simulation Environment

Browser-Based Educational IDE

A dual-editor VPython simulation environment combining a Blockly visual block editor with direct GlowScript VPython code editing, and a live isolated 3D viewport. Designed for teaching physics through interactive simulation.

Key features: - Blockly-based visual physics modelling – no-code simulation building - Direct VPython code editing with live 3D rendering - Prebuilt templates: projectile motion, orbital mechanics, and more - Custom blocks for cleaner animation chains - Dark/light mode with viewport text readability - Export support for sharing simulations

Technology Stack: [JavaScript](#) · [Blockly](#) · [GlowScript VPython](#) · [Node.js](#) · [HTML/CSS](#)

Anti-Temu – Chrome Extension

March 2025

A privacy-preserving Manifest V3 Chrome extension that permanently hides distracting ads, popups, and elements on any webpage – with smart ad detection, per-site persistence, element picker, undo support, and a fully themed popup UI.

Key features: - Smart ad/element detection with MutationObserver for dynamically injected content - Permanent per-site element hiding with localStorage persistence across sessions - Interactive element picker with visual highlighting before hiding - Full undo/redo history for hidden elements - Theme system (dark/light) with a clean popup management interface - Zero external dependencies – pure Manifest V3 architecture

Technology Stack: [Chrome Extensions \(Manifest V3\)](#) · [Service Workers](#) · [Content Scripts](#) · [JavaScript](#) · [HTML](#) · [CSS](#)

Hangman – Command-Line Game

April 2025

GitHub: github.com/DrVanHelsing/Hangman

A clean, minimal Hangman game implementation with word list loading, difficulty selection, and round-based scoring – built as a focused, well-structured learning project.

Tic Tac Toe

March 2025

GitHub: github.com/DrVanHelsing/TicTacToe

A classic Tic Tac Toe game featuring two-player local gameplay and optional Human vs Computer mode. Focused on clean code structure and beginner-friendly game logic implementation.

Fitness Center Management System

University of the Western Cape

A relational database management system with a full web front-end. Designed and implemented in MySQL with a PHP, HTML, and JavaScript interface for complete CRUD operations.

Links: [Project Report](#) · [Demo Video](#)

Technology Stack: [MySQL](#) · [PHP](#) · [HTML](#) · [CSS](#) · [JavaScript](#) · [SQL](#)

// Personal Traits

- Curious and self-motivated learner
 - Strong interest in AI, machine learning, and emerging technologies
 - Passionate about education, mentoring, and knowledge transfer
 - Enjoys team collaboration, hackathons, and innovation challenges
 - Lifelong learner with a continuous growth mindset
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// Online Portfolio

Scan the QR code or visit <https://www.iamtredir.com> to explore the full portfolio.



<https://www.iamtredir.com>

Built with [React](#) · [Vite](#) · Deployed on [Vercel](#)
